



Course Code: EEE373
Course Title: Embedded System Design

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Section: 02

Assignment

4. Here, ID: 21221030

last digit $Y = 0$.

$$\therefore f = 1000 \text{ Hz}$$

$$T = \frac{1}{f} = \frac{1}{1000} = 10^{-3} \text{ secs}$$

(a) Given frequency of AT Mega 32 is, $f = 12 \text{ MHz}$

$$\therefore \frac{1200 \times 10^4}{\text{sec}} \times 1 \times 10^{-3} \text{ sec} \\ = 12000 \text{ cycle}$$

So to count 10^{-3} secs it needs 12000 cycles

Where as, Timer0 has, $2^8 - 1 = 255$ cycles.

$$\text{A prescaler of } 1024 = \frac{1024}{12 \times 10^6}$$

$$= 8.53 \times 10^{-5} \text{ secs}$$

In a square wave, the duty cycle gets half = $\frac{10^{-3}}{2}$

$$= 5 \times 10^{-4} \text{ secs}$$

$$\text{Hence, for half duty cycle, } = \frac{5 \times 10^{-4}}{8.53 \times 10^{-5}}$$

$$= 5.86 \approx 6 \text{ pulses}$$

Code:

```
#include <mega32.h>
```

```
interrupt [TIMO_COMP] void timer0_comp_isr (void)
```

```
{
```

```
PORTA.0 = ~ PORTA.0 ; // toggles state.
```

```
TCNT0 = 0; // initializes Timer 0 at 0.
```

```
}
```

```
void main (void)
```

```
{
```

```
DDRA.0 = 1;
```

```
TCCR0 = (0 << WGM00) | (0 << COM01) | (0 << COM00) |
```

```
(0 << WGM01) | (1 << CS02) | (0 << CS01) |
```

```
(1 << CS00); // Prescaler 1024 is set.
```

```
TCNT0 = 0;
```

```
OCR0 = 5; // Compare at 6th pulse
```

```
TIMSK = 0b00000010;
```

```
#asm ("sei") // Enables global interrupt
```

```
PORTC.0 = 0;
```

```
while(1)
```

```
{ // runs infinite loop
```

```
}
```


(b) without using any interrupt,

50% duty cycle gives = 0.5 ms [from 'a']

So, here we can toggle PORTA.0 at every 0.5 ms.

Code: #include <mega32.h>
#include <delay.h>

```
void main ()  
{  
    DDRA.0 = 1;  
    while (1)  
    {  
        PORTA.0 = ~ PORTA.0; // state changes at every  
        delay_ms (0.5) // 0.5 ms (high for 0.5 ms and low  
                        // for 0.5 ms)  
    }  
}
```


2. Given, frequency of PWM wave = 75 Hz

$$\therefore T = \frac{1}{f} = \frac{1}{75} = 0.0133 \text{ secs}$$
$$= 13.33 \text{ ms}$$

$$\text{At 60\% duty cycle} = 60\% \times 13.33$$
$$= 8 \times 10^{-3} = 8 \text{ ms.}$$

$$\text{A crystal with } 12\text{MHz} = \frac{12 \times 10^6}{\text{secs}} \times 8 \times 10^{-3} \text{ secs}$$
$$= 96000 \text{ cycles.}$$

Using Timer 2 with prescaler 1024 gives

$$f = \frac{12 \times 10^6}{1024} = 11718.75 \text{ Hz}$$

$$\therefore T_2 = \frac{1}{f} = 8.53 \times 10^{-5} \text{ secs}$$

60% of duty cycle
for that we need,

$$= \frac{8 \times 10^{-3}}{8.53 \times 10^{-5}}$$

$$= 93.75$$

$$\approx 94 \text{ pulses.}$$

$$\text{It remains high for } 8 \text{ ms and low for } = (13.33 - 8) \text{ ms}$$
$$= 5.33 \text{ ms}$$

$$\therefore \text{Now, we need, } = \frac{5.33 \times 10^{-3}}{8.53 \times 10^{-5}} = 62.52$$
$$\approx 63 \text{ pulses.}$$

Code: #include <mega32.h>
#include <delay.h>

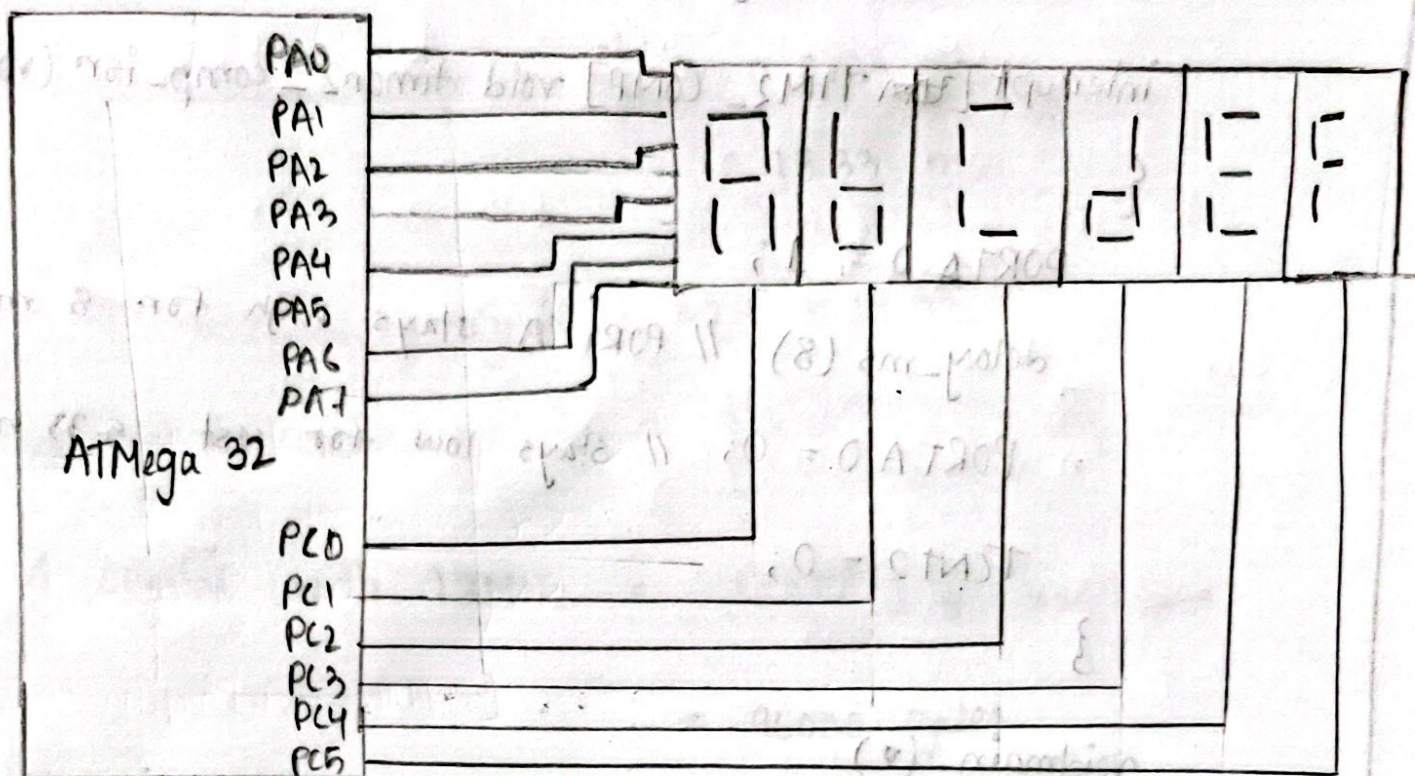
interrupt [~~TM~~ TIM2_COMP] void timer2_comp_isr (void)
{
PORTA.0 = 1;
delay_ms (8) // PORT A stays high for 8 ms
PORTA.0 = 0; // stays low for rest 5.33 ms.
TCNT2 = 0;
}

void main ()

{
DDRA.0 = 1;
ASSR = 1 << AS2; // External ^{crystal} is being used.
TCCR2 = (1 << CS22) | (1 << CS21) | (1 << CS20); // Prescaler of 1024 is set
TCNT2 = 0;
OCR2 = 0x3E; // Compares and raises flag at 63th pulse,
TIMSK = 1 << OCIE2; // hex code for 62 is given
#asm ("sei") // Enables global interrupt
PORTA.0 = 0; // initialized at 0.
while (1)
{
// runs infinite loop
}
}

3.

Circuit diagram:



To generate ABCDEF:

$L = C.A200 \left\{ \begin{array}{c} a \\ \hline b \\ \hline c \end{array} \right\}$ (7 segment)
pin

hex code

0 1 1 1 0 1 1 1 $\rightarrow 0_{x} 77$

$0 \rightarrow 0x7C$

0 0 1 1 1 0 0 1 $\rightarrow$ 0x39

0 0 1 0 1 1 1 1 0 $\rightarrow 0x5D$

$$0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad \Bigg| \begin{array}{l} - \\ - \\ - \end{array} \rightarrow 0x79$$
$$0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 1 \quad \begin{array}{c} \text{---} \\ | \end{array} \rightarrow 0x71$$


```

Code: #include <mega32.h>
#include <stdio.h>
#include <delay.h>

```

```

void main ( )

```

```

{

```

```

    DDRA = 0x FF; // Data port A

```

```

    PORT DDRC = 0x FF; // Control port C

```

```

    while (1)
    {

```

```

        PORT C = 0b11111110; // LED output in seven segment

```

```

        PORTA = 0x 77; // Sets A

```

```

        delay-ms (10);

```

```

        PORTC = 0b11111101;

```

```

        PORTA = 0x 7C; // Sets b

```

```

        delay-ms (10)

```

```

        PORTC = 0b11111011;

```

```

        PORTA = 0x 39; // Sets C

```

```

        delay-ms (10)

```

```

        PORTC = 0b11110111;

```

```

        PORTA = 0x 5D; // Sets d

```

```

        delay-ms (10)

```

```

        PORTC = 0b11101111;

```

```

        PORTA = 0x 79; // Sets E

```

```

        delay-ms (10)
    }
}

```



```

PORTC = 0b11011111; <1.58 0p0m> abubri # : 0b00
PORTA = 0x71; // sets <f.0ibte> abubri #
delay_ms(10) <1.58 0p0m> abubri #
}

```

```

}

```

4. #include <mega32.h>

#include <stdio.h>

#include <delay.h>

interrupt [EXT_INT0] void ext_int0_isr(void)

```

{

```

PORTA = 0xFO;

delay_ms(20000) // interrupts for 20 seconds when

```

}

```

switch is pressed.

void main()

```

{

```

DDRA = 0xFF;

GICR = (0<<INT1) | (1<<INT0) | (0<<INT2) // Enables INT0.

MCUCR = (0<<ISC11) | (0<<ISC10) | (1<<ISC01) | (0<<ISC00)

// generates interrupt when INT0 changes

#asm("sei") // Enables global interrupt.

while(1) // infinite loop generation

```

{

```

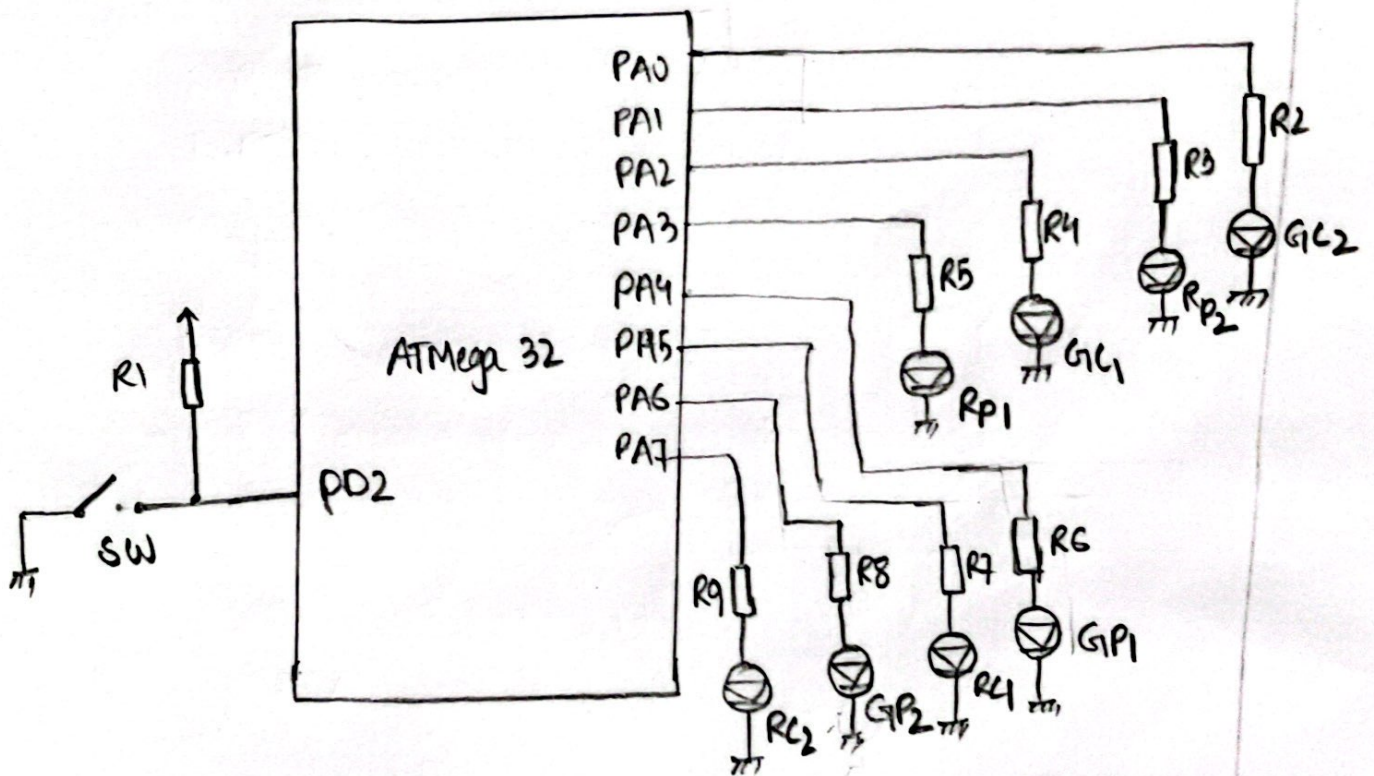
PORTA = 0x0F; // normal LED operation

delay-ms (45000); // 45 seconds ON for GL1, GL2, RP1, RP2

PORT A = 0x F0;

delay-ms (15000) // 15 seconds ON for RL1, RL2, GP1, GP2

Circuit for this system:



Here,

GL1, GL2, GP1, GP2 → set as LED-GREEN

RP1, RP2, RL1, RL2 → set as LED-RED.